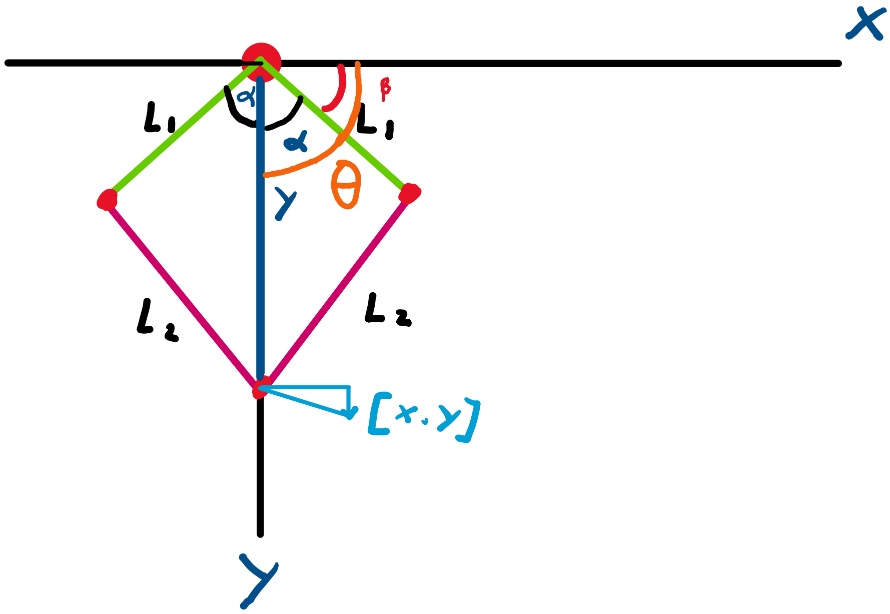
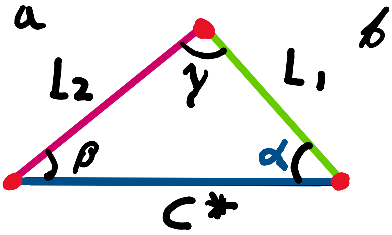



(Voor hoek alpha α)

- Cosinusregel $\qquad c = \sqrt{x^2 + y^2}$

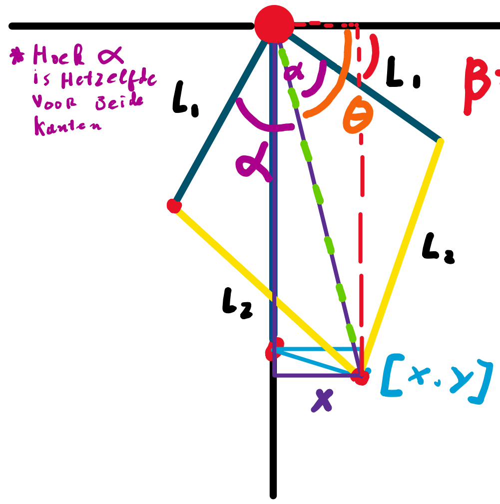



$$a^2 = b^2 + c^2 - 2 \cdot bc \cdot \cos(\alpha)$$
$$\downarrow$$
$$L_2^2 = L_1^2 + c^2 - 2 \cdot L_1 \cdot c \cdot \cos(\alpha)$$
$$\downarrow$$
$$L_2^2 - L_1^2 - c^2 = -2 \cdot L_1 \cdot c \cdot \cos(\alpha)$$
$$\downarrow$$
$$\frac{L_2^2 - L_1^2 - c^2}{-2 \cdot L_1 \cdot c} = \cos(\alpha)$$
$$\downarrow$$
$$\boxed{\alpha = \cos^{-1}\left(\frac{L_2^2 - L_1^2 - c^2}{-2 \cdot L_1 \cdot c}\right)}$$

* Hoek α is Hetzelfde .

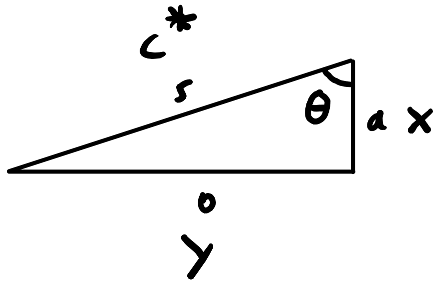

$\beta = \theta - \alpha$

*




$\beta = \theta - \alpha$

Voor Hoek θ ⟹




- Geeft:

$\theta = \tan^{-1}\left(\frac{o}{a}\right) = \tan^{-1}\left(\frac{y}{x}\right)$

↓

$$\boxed{\begin{aligned} \theta &= \tan^{-1}\left(\frac{y}{x}\right) \\ \beta &= \theta - \alpha \end{aligned}}$$

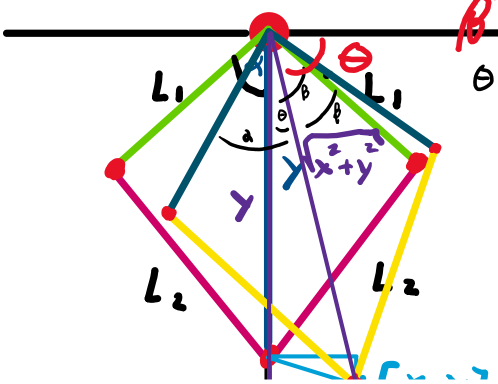
Afmetingen 4-BAR

- $L_1$ : 76,8 mm
- $L_2$ : 120 mm

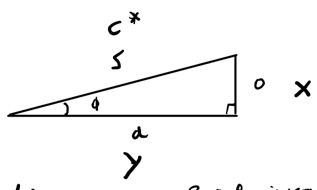
$\alpha = \alpha -$

$\alpha = \ldots\ldots$

$\beta =$

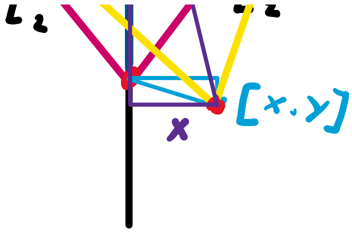



$\theta =$




Liefst geen Berekeningen

$\theta = \tan^{-1}\left(\frac{x}{y}\right)$




$\theta = \tan^{-1}\left(\frac{x}{y}\right)$

7 Hoek $\beta = \theta + \alpha$

. Hoek $\alpha = \theta - \alpha$